

LFP Lithium Battery System For Communication

User's Manual

Shuangdeng Group Co.,Ltd

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1、 Brief introduction

LFP Lithium battery system for communication is a high-tech product, it's manufactured and developed successfully by Shuangdeng Group ; The product has many merits mainly includes: integration, miniaturization, light-weight, intelligence centralism and monitoring , maintenance and management, unattended, standardization installation and easy operation, energy conservation ,environmental protection. Now the product have been widely used in backup power of telecommunication field such as access network equipment, far-end telephone exchange,mobile telecommunication equipment, transmission facility, satellite earth station and microwave communication equipment etc. At present , Shuangdeng Group's LFP backup battery telecommunication series products are volume production and be widely used in the domestic telecommunication field.

2、 Production feature

- (1) Battery's positive pole using LiFePO_4 material with high cycling life and good security feature;
- (2) The battery system using high performance battery management system(BMS),the BMS have the current, voltage, temperature protect function , and make the battery system have a good telecommunication with the host machine;
- (3) supervision unit automatic measure the battery's charge and discharge current, voltage, cell's surface temperature and environment temperature.
- (4) Secondary knock off function, when the battery's voltage lower than the alarm value there will be alarm signal ,when the battery's voltage too low the battery will automatic knock off, to protect the battery
- (5) The battery system has a good electromagnetic compatibility;
- (6) with entire intelligent design, the disposition with centralism monitoring module, and it has a four remote function,they are telesignalisation, remote regulating , remote control and remote debugging, and be managed by computer , it also can communication with the remote central control center .
- (7) The power source control technology combined with computer and can monitor and control each kind of parameter and condition at each time;
- (8) With the self-cooling method, the system have very low noise.

3、 Production advantage

- (1) The product can directly working in a constant charge and discharge working mode at the original dc telecommunication switching power supply system .
- (2) LFP backup battery system is the high-tech product manufactured by Shoto Group, with proprietary intellectual property rights, the product fill in the blank inland,and the key technical indexes are in the lead level of the world.
- (3) Shuangdeng Group's LFP lithium backup battery system are quantity production inland,and be widely used in the telecommunication backup battery field,with the longest market time, biggest market quantity and good market feed back ;

- (4) In order to satisfy Telecom Operator's new demand of the power-supply system, Shuangdeng Group made the solution upon the LFP battery 、 BMS、 DC switch power module(or ups uninterrupted power system)
- (5) Higher density per unit volume and gravimetric specific energy,during installing, there is no special demand for space and weight bearing, greatly reduce the cost of renting area
- (6) LFP lithium cell's cycle life are more than 2000 times
- (7) Good temperature characteristic : working environment temperature can reach - 20 ~ + 60℃ (recommended temperature : -15 ~ + 50℃) , greatly reduce the equipment acquisition cost and power consumption cost.
- (8) LFP lithium cell's structure is leakproof structure, it can be put any where with no harmful gas release , non-pollution, it's a green product.
- (9) LFP lithium battery has very good rate discharge performance, low capacity LFP battery system can adapt to the large current discharge occasion.
- (10)Allocation flexible:since multi LFP lithium battery modules in parallel, the high capacity of LFP lithium battery system become true, not only enhance the the system's output power, but also prolong the the system's reserve time .

4、 Product type module and technical parameter

Chart 4-1 48V series of product and model

Model	Nominal voltage (DC V)	Nominal capacity (Ah)	maximum load current (A)	dimension (mm) (wide×height×deep) (do not include the handle)	weight (kg)±2kg	remarks
SDA10-4810	48	10	7.9	442×44×240 (1U)	8.5	Can in parallel
SDA10-4810D	48	10	13.5	442×44×240 (1U)	8.5	Can in parallel
SDA10-4820	48	20	20	442×88×300 (2U)	16.2	Can in parallel
SDA10-4830	48	30	30	442×133×300 (3U)	21.5	Can in parallel
SDA10-4840	48	40	40	442×177×300 (4U)	27.5	Can in parallel
SDA10-4850C	48	50	40	442×177×300 (4U)	31.8	Can in parallel
SDA10-4850	48	50	50	442×133×380 (3U)	31.8	Can in parallel
SDA10-4850G	48	50	50	442×133×520 (3U)	39	Can in parallel
SDA10-48100	48	100	50	Two 4850 in parallel	--	parallel system
SDA10-48150	48	150	75	Three 4850 in parallel	--	parallel system
SDA10-48200	48	200	100	Four 4850 in parallel	--	parallel system
SDA10-48250	48	250	125	Five 4850 in parallel	--	parallel system
SDA10-48300	48	300	150	Six 4850 in parallel	--	parallel system
SDA10-4810-I	48	10	3	442×44×455 (1U)	10.5	AC in DC out

Chart 4-2 24V series of product and model number

Model	Nominal voltage (DC V)	Nominal capacity (Ah)	maximum load current(A)	dimension (mm) (wide×height×deep)	weight (kg)	remarks
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SDA10-2450	24	50	50	442×88×300(2U)	15.5	Can in parallel
SDA10-24100	24	100	50	Two 2450 in parallel	31	parallel system
SDA10-24150	24	150	75	Three 2450 in parallel	46.5	parallel system
SDA10-24200	24	200	100	Four 2450 in parallel	62	parallel system
SDA10-24250	24	250	125	Five 2450 in parallel	77.5	parallel system
SDA10-24300	24	300	150	Six 2450 in parallel	93	parallel system

Remarks : 1. The battery system supports parallel connection to extend back-up time, but the maximum load current will not be multiplied with the parallel quantity. The current parallel system can reach capacity of 48V2000Ah, and more capacity can be customized but longer time required accordingly.

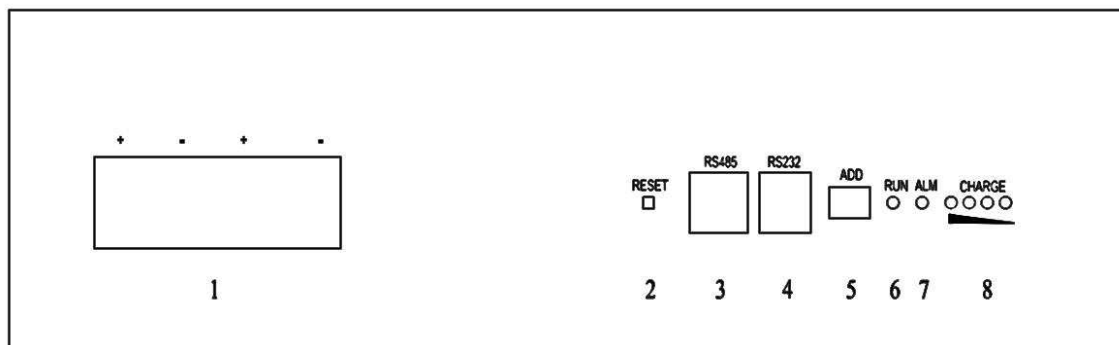
2. The dimension and weight of the parallel system is for battery, not including cabinet or rack.

3. Please configure the parallel system correspondence with the load, if more discharge current required, please contact with the manufacturer to ask for customized product if available.

5、Using environmental requirements

- Ambient temperature : -20 ~ +60℃ (suggest temperature : -15-+50℃) ;
- Storage Temperature : -40 ~ +80℃ ;
- Relative humidity : 5% ~ 95%RH ;
- Height : not exceeding 4000m ;
- No conductive dust and corrosive gas place.

6、System case panel function



6.1 Case panel schematic drawing for the case height is 2U or 4U system

6.2 Battery output terminal

“1” shows the output port of the battery , Specification models are 24V、 48V batteries use 4 cores binding post , The main consideration is to make battery modules can be used in cascading situation , 4 pins of the biding post from left to right are defined as battery+、 battery-、 battery+、 battery-.

Chart 6-1 different specification of batteries , instruction of the output port wiring configuration

Battery model	Battery capacity	Output crew specification	Wiring diameter request	wire nose
SDA10-4810	10AH	M4	2.5mm ²	2.5-5
SDA10-4820	20AH	M6	≥4mm ²	4-6
SDA10-4830	30AH	M6	≥4mm ²	4-6
SDA10-4840	40AH	M6	≥6mm ²	6-6
SDA10-4850	50AH	M6	≥10mm ²	10-6
SDA10-48100	100AH	--	≥10mm ²	10-6
More than 100AH	--	--	X (X≥10mm ²)	X-6

6.3 Reset key

“2”“RESET”reset key。 When the system’s operation is abnormal ,can use the key to reset the system , ensure the stability of the system。

6.4 Communication interface

6.4.1 Cascade communication interface

“3”RS485 communication interface。 When the system is in cascade , the way of data transmission is the RS485 serial communication。 Chart 6-1 is the RS485 port definition list。 After cascading the inner of the Pack system in the RS485 communication pattern , upper computer system get the data from each SlavePack from the MasterPack。

Chart 6-2RS485 port define

pin	Define instruction
1、 8	RS485-B
2、 7	RS485-A
3、 6	earth
4、 5	NC (hang in the air)

6.4.2Connect the communication interface

“4”RS232 communication interface.When system uploading data select the RS232 serial port communication method and table 5-2 is RS232 interface definition.The transmission data content includes:system parameter,system state and alarm information.When use RS232 to communicate, the baud rate should be set as 1200bps.The system ‘s dials code switch should be set as Pack1 (master pack) mode, then the RS232 communication module will be valid , to communicate with the position machine.

Chart 6-3 RS232 interface definition

6cores Crystal head pin	Definition explanation	DB9 needle pin	Definition explanation
2	Earth		
3	RX	3	TX

4	TX	2	RX
5	Earth	5	Earth

6.4.3 Communication wire

Picture 6-2 is the upload communication wire of RS232 ; Picture 6-3 is the explanatory drawing of RS232's communication wire's crystal's first and sixth port without connection.

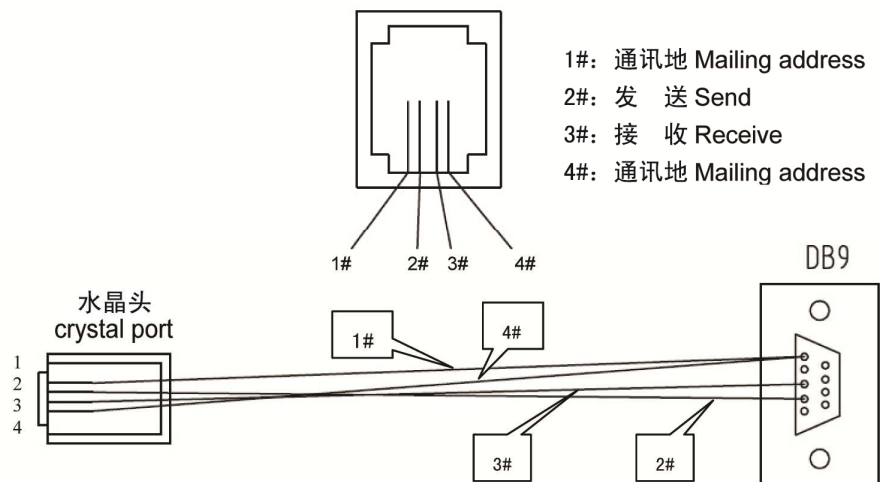
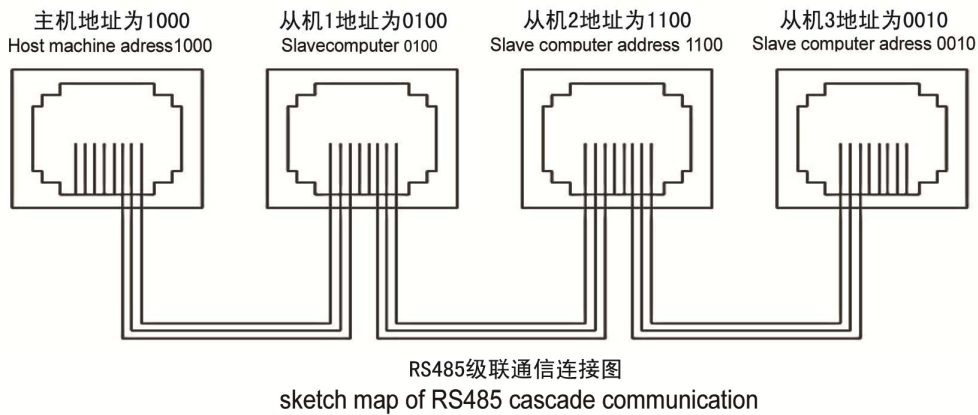


图6-2 RS485级联通信连线和RS232上传数据通信连线

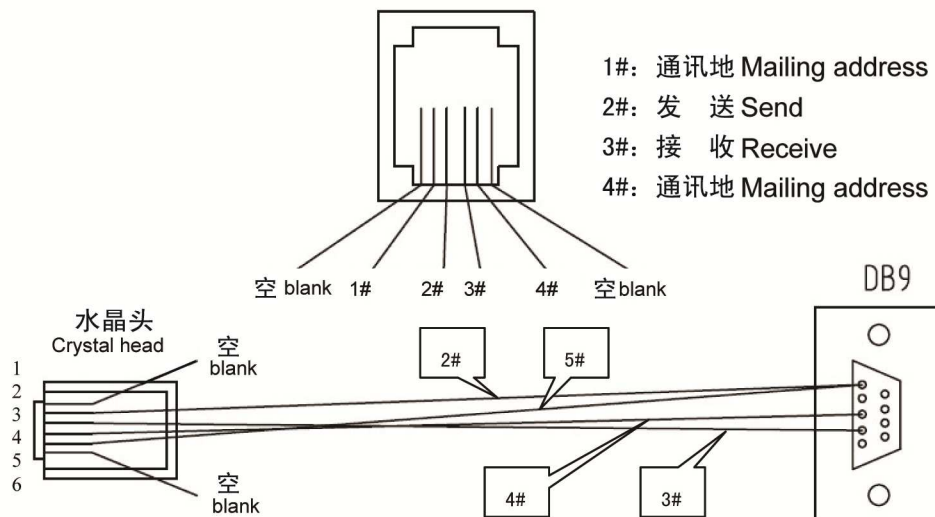


图6-3 RS232通信连线的六芯水晶头中第一和第六个焊片为空

Picture 6-3 RS232 communication wires's crystal port's first and sixth soldering lugs are blank

6.5 Dial switch

“5”represent the dial switch , on the panel there is a 4 numerals dial switch which can be used to setting the model’s address and table 6-4 is a operation instruction.

Picture 6-4 is the wave-band address code setting chart

Wave band				Address	Pack definition	Definition explanation
1	2	3	4			
ON	OFF	OFF	OFF	0	Pack1	As Master Pack,permit RS232 communication
OFF	ON	OFF	OFF	1	Pack2	As Slave Pack1
ON	ON	OFF	OFF	2	Pack3	As Slave Pack2
OFF	OFF	ON	OFF	3	Pack4	As Slave Pack3
ON	OFF	ON	OFF	4	Pack5	As Slave Pack4

6.6 System running indicator light

“6” the system running indicator light : RUN light ,green. Bright during charging ,during discharging 1second bright , and1second extinguish.

6.7 Alarm indicator light

“7” the Alarm indicator light : ALM light,red. When the system have fault it will bright.。

6.8 Capacity signal light

“8” the capacity light, 4 green light , every light means 25% capacity.When capacity is 100% , 4 lights are all bright; when capacity is 75% , left 1 light extinguish, and right 3 lights bright; when capacity is 50%, left 2 lights extinguish, and right 2 lights bright; when capacity is 25%, left 3 lights extinguish, and right 1light bright.

6.9 LED signal light introduction

Table 6-5 LED signal introduction

System state	Operation state	RUN	ALM	LED				Explanation
								
Power off	Dormancy	ext	ext	ext	ext	ext	ext	All extinguish
Readiness for action	normal	Shi1	ext	ext	ext	ext	ext	Readiness for action
	Alarm	ext	Shi 3	ext	ext	ext	ext	low voltage
Charge	normal	Bri	ext	According capacity to signal				According capacity shi2
	protect	Bri	ext	Bri	Bri	Bri	Bri	Stop charge
discharge	normal	shi3	ext	According capacity to signal				According capacity to bright signal
	Alarm	shi3	shi3					
	Over current protect	ext	Bri	ext	ext	ext	ext	Stop discharge
	Low voltage protect	ext	ext	ext	ext	ext	ext	Stop discharge
temperature	normal	According to the condition normally to indicate						---
	protect	ext	Bri	ext	ext	ext	ext	---

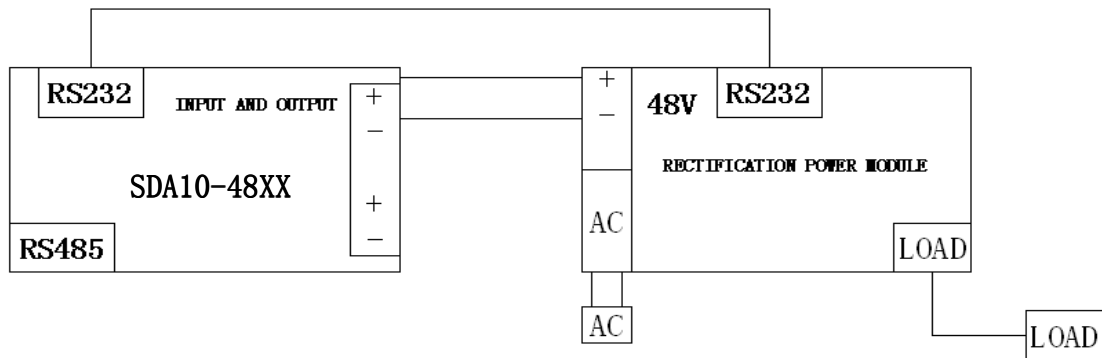
Table 6-6 LED shining specification

Shinning style	Bright	Extinguish
shi1	0.25s	3.75s
shi2	0.5s	0.5s
shi3	0.5s	1.5s

7、System working principle and parameter

7.1 System working principle

SDA10-48xx and SDA10-24xx LFP lithium backup battery system working principle like chart 7-1 shows. AC 220V input the battery system ,and be disposed by the rectifier power source module , output -48V or -24V. when the electrical network normal, the rectifier power module supply the load working power and charging the backup battery system at the same time; when electrical network cut-off, the backup battery system supply the direct current to ensure direct-current power supply system operate normally, which same as the uninterrupted power supply function. The power supply have the secondary knock off function. when the battery system's voltage excessively low, the battery management system will automatic knock off the battery's connection, to protect the battery's life.



Picture 7-1LFP lithium backup battery system working principle

7.2 Charge parameter

Chart 7-1 series products charge parameter

Model	Charge voltage (DCV)			Charge current (A)		
	Minimum	Typical value	Maximum	Min value	Limits value	Maximum
SDA10-2450	29.2	30.4	30.8		5.0	
SDA10-4810	53.2	56.4	57.5		2.0	
SDA10-4820	53.2	56.4	57.5		5.0	
SDA10-4830	53.2	56.4	57.5		5.0	
SDA10-4840	53.2	56.4	57.5		5.0	
SDA10-4850C	53.2	56.4	57.5		10.0	25.0

SDA10-4850 (3U)	53.2	56.4	57.5		5.0	10.0
SDA10-4850G	53.2	56.4	57.5		10.0	25.0
SDA10-48100	53.2	56.4	57.5		20.0	50.0
SDA10-48150	53.2	56.4	57.5		30.0	75.0
SDA10-48200	53.2	56.4	57.5		40.0	100.0
SDA10-48250	53.2	56.4	57.5		50.0	125.0
SDA10-48300	53.2	56.4	57.5		60.0	150.0

Remarks: before using the battery, make sure the charging voltage is not lower than the least value in the chart, to avoid the battery can't be fully charged, then the discharge capacity not enough.

Lithium battery capacity of more than 100AH, the default using 4850C parallel.

7.3 Discharge parameter

Chart 7-2 series of products' discharge parameter

Model	Discharge voltage (DCV)			Discharge current (A)		
	Minimum	Typical value	Maximum	Minimum	Typical value	Maximum
SDA10-2450	20	24.0	27.0		25	50
SDA10-4810	40	48.0	55.0		5	7.9
SDA10-4810D	40	48.0	55.0		5	13.5
SDA10-4820	40	48.0	55.0		10	20
SDA10-4830	40	48.0	55.0		15	30
SDA10-4840	40	48.0	55.0		20	40
SDA10-4850C	40	48.0	55.0		25	40
SDA10-4850(3U)	40	48.0	55.0		25	50
SDA10-4850G	40	48.0	55.0		25	50
SDA10-48100	40	48.0	55.0		30	50
SDA10-48150	40	48.0	55.0		45	75
SDA10-48200	40	48.0	55.0		60	100
SDA10-48250	40	48.0	55.0		75	125
SDA10-48300	40	48.0	55.0		90	150

7.4 System protection function and parameter

Chart 7-324V battery system all kinds of protection function and parameter

Parameter name	Protection function	24Vsystem protection parameter	48Vsystem protection parameter	Dropout value
Voltage	Cell overvoltage protection	3.85V	3.85V	Capacity of the drop value ≥4 % release

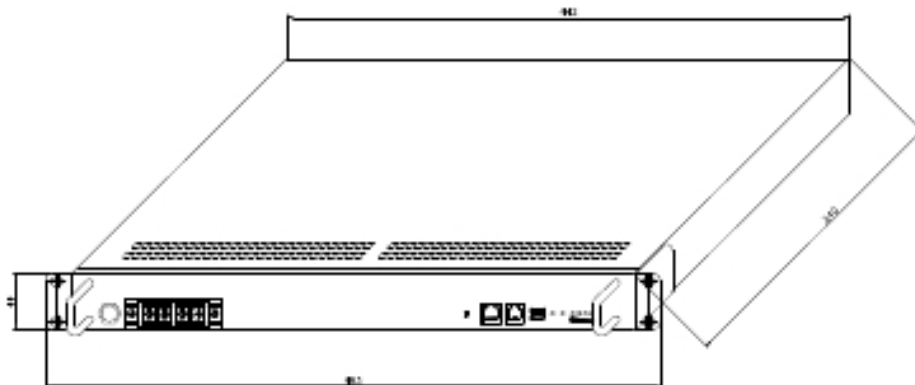
Parameter name	Protection function	24Vsystem protection parameter	48Vsystem protection parameter	Dropout value
	Cell undervoltage alarm	2.8V	2.8V	/
	Cell voltage low protection	2.5V	2.5V	Charge release
	Total low voltage alarm	22.4V	42.0V	/
	Total undervoltage protection	20.0V	40.0V	Charge release
Current	Discharge over current protection	53A	1.1C	After 30s release
	Charge current limit value	0.1C	0.1C OR 0.2C	/
	short-circuit protection function	yes	yes	Release after 30s
Temperature	Charge high temperature protection	65℃±2℃	65℃±2℃	60℃±2℃recover
	Charge low temperature protection	-20℃±2℃	-20℃±2℃	15℃±2℃recover
	Discharge high temperature protection	65℃±2℃	65℃±2℃	60℃±2℃recover
	Discharge low temperature protection	-25℃±2℃	-25℃±2℃	-20℃±2℃recover

Remark: because the product's parameter update occasionally, some models of battery have difference with the chart above, Please follow the specifications , There's no further notice from the manufacturer.

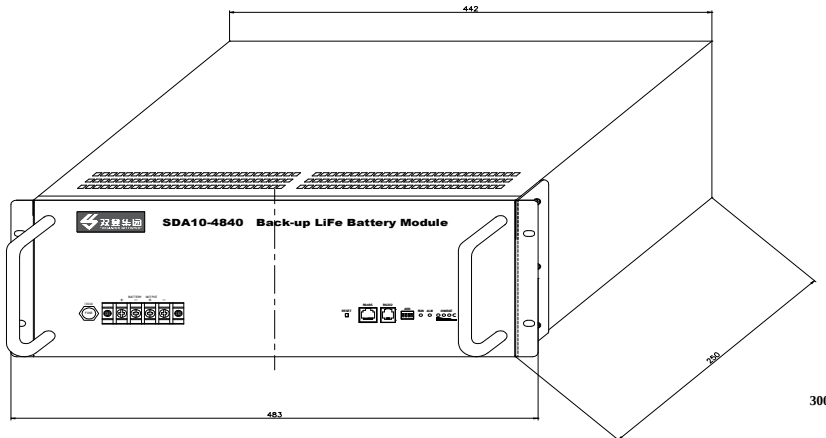
8、 system installment 、 use and maintain

8.1 System installment

8.1.1 Height of 1U 、 3Uand 4U system's installment



Picture 8-1 height of 1U system's structure dimensional drawing



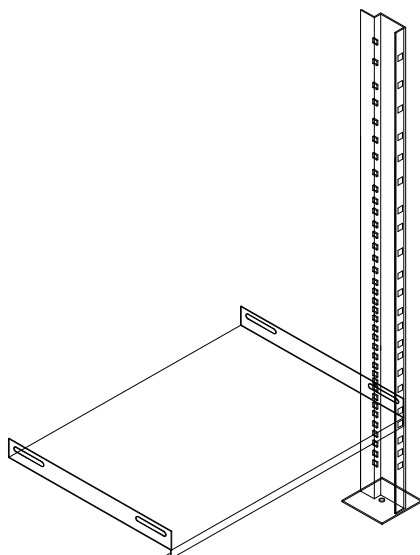
Picture 8-2 height of 4U system's structure dimensional drawing

- (1) We recommend installs in the 19" standard cabinet, it also can be installed in the wall way of hanging on the wall, Use 4 M10 bolts to fix the system case 's both sides' ears on the rack;
- (2) From the case's panels' right earth connection hole, use 2.5mm² black flexible wire to connect the earth and make sure earth connection well;
- (3) When the LFP lithium battery system's capacity under 20AH , Use thick more than 2.5mm² red and black flexible cord to connect LFP lithium backup battery system's output terminal post(“ + ”,“ - ”) to the switching power module's positive post and negative post.when the LFP lithium battery system's capacity above 20AH , use thick more than 4mm² red and black flexible wire to connect LFP lithium backup battery system's output terminal (“ + ”,“ - ”) to the switching power module's positive post and negative post. Pay attention to the positive and negative mark.

8.1.2 HSNF installment

Height 4U is the overall dimension of 40Ah and 50Ah LFP communication battery system ,can in parallel connection as the chart 4-1 and 4-2 shows : 24V and 48V series of product specification .the way to install the system is just below :

- (1) The installment of the HSNF 50Ah LFP communication battery system

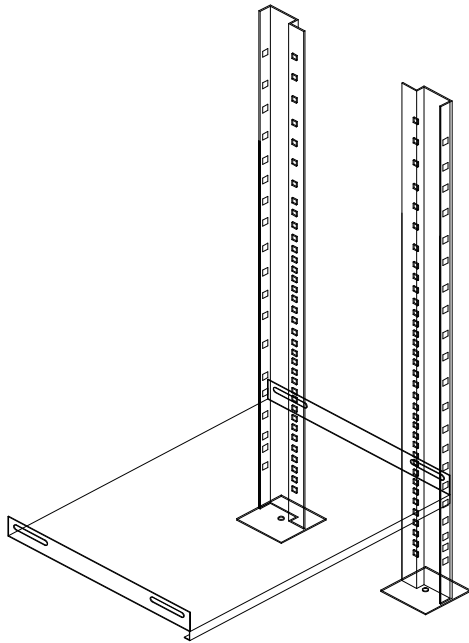


First step :

instruction : As the picture 8-3 showing, install the tray and 1# holder first.

Make sure both of them not vertical.Make sure the distance between plane of the tray and surficial is 180mm

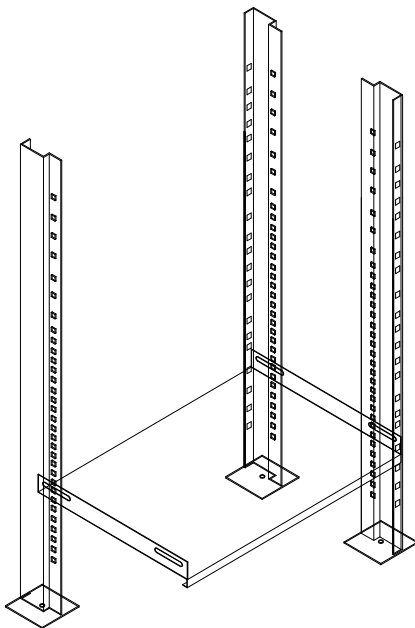
Picture 8-3 1# the sketch map of HSNF installment



Step two :

instruction : As picture 8-4 shows , install holder 2# ,
make sure holder2# and holder 1# are keep parallel

Picture 8-4 2# the sketch map of HSNF installment.



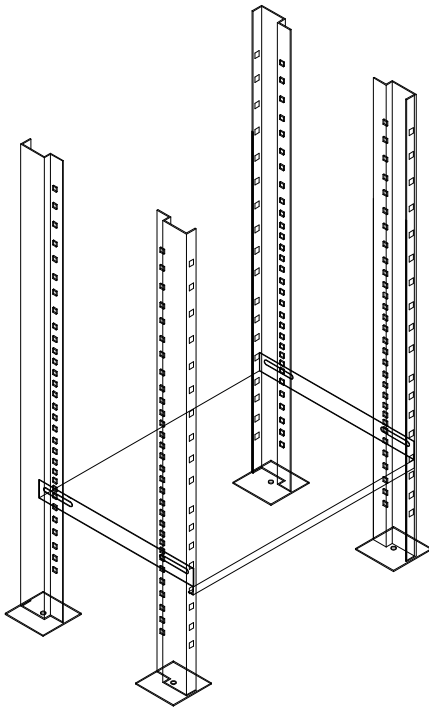
Step three :

Instruction : As the picture 8-5 shows , install 3#
holder , make sure holder3# and holder 1# are keep parallel.

Picture 8-5 3# the sketch map of HSNF installment

Step four :

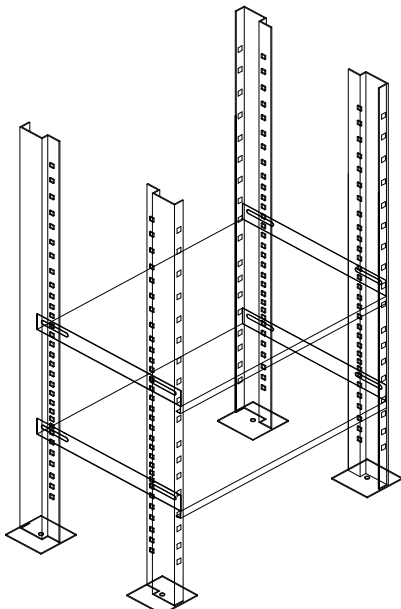
Instruction : As picture 8-6 shows , install holder 4# , make sure holder 4# and holder 1# are keep parallel.



Picture 8-6 4# HSNF installment sketch map

Step five :

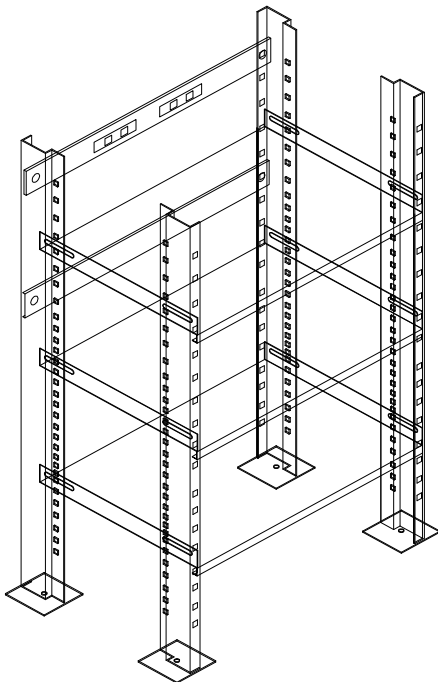
Instruction :As picture 8-7 shows install the second tray ,pay attention that the distance to the first tray is 220mm. Make sure the second tray and the first tray in parallel



Picture 8-7 sketch map of installing the second tray

Step six :

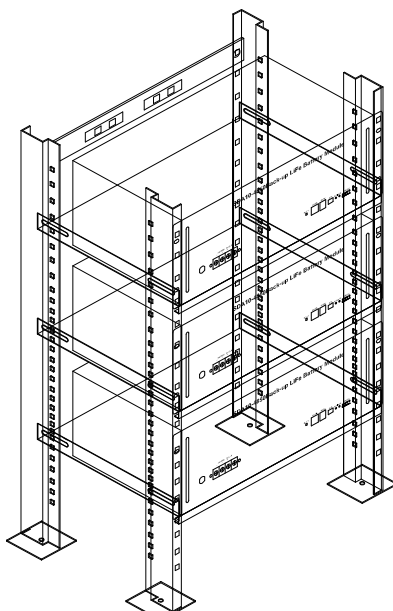
Instruction : As picture 8-8 shows , install the third tray, the distance between the second and third tray is 220mm. Make sure the third tray and the first tray in parallel. Install busbar and baffle. Then use expansion screw to fix the whole holder on the floor.



Picture 8-8 sketch map of installing the insulator fix strip and confluence terminal

50Ah

Picture



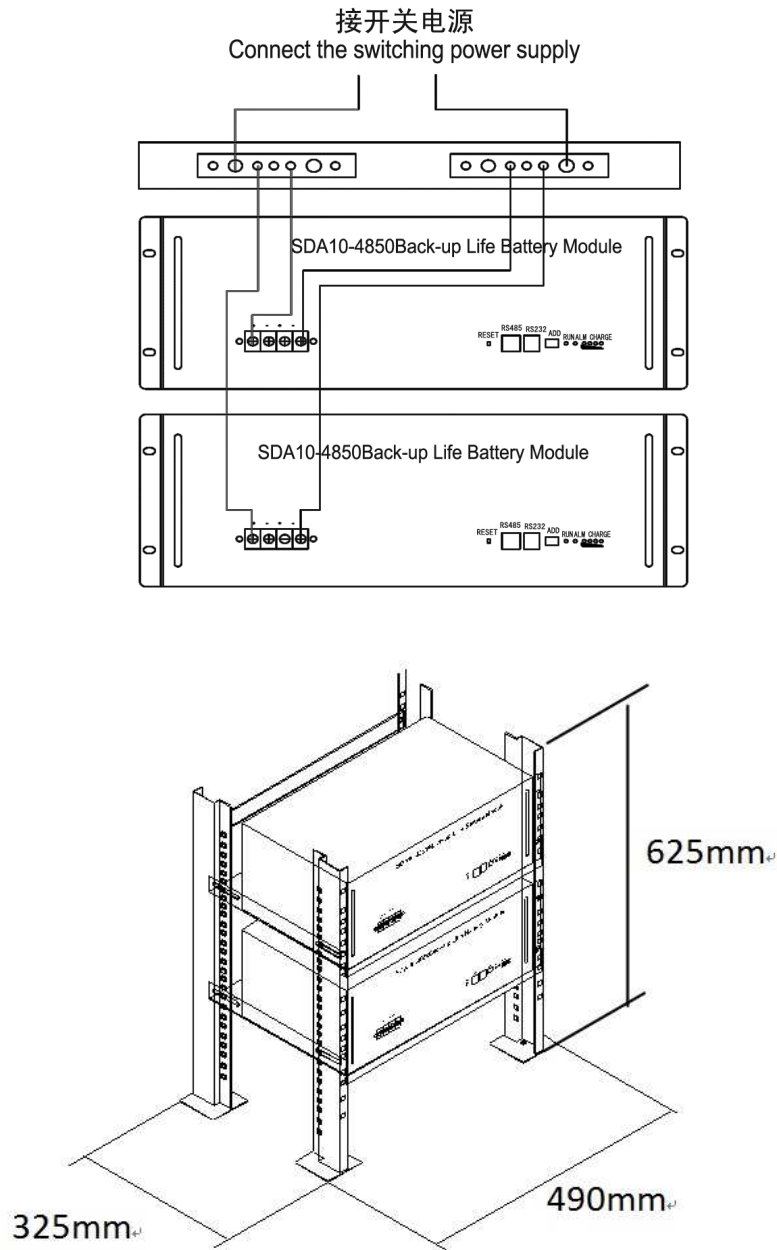
Step seventh :

Instruction : As the picture 8-8shows , fix the LFP lithium battery on the holder

8-9 is a sketch map of installing 50Ah LFP lithium communication battery

(2)The installation of HSNF 100Ah LFP communication battery system

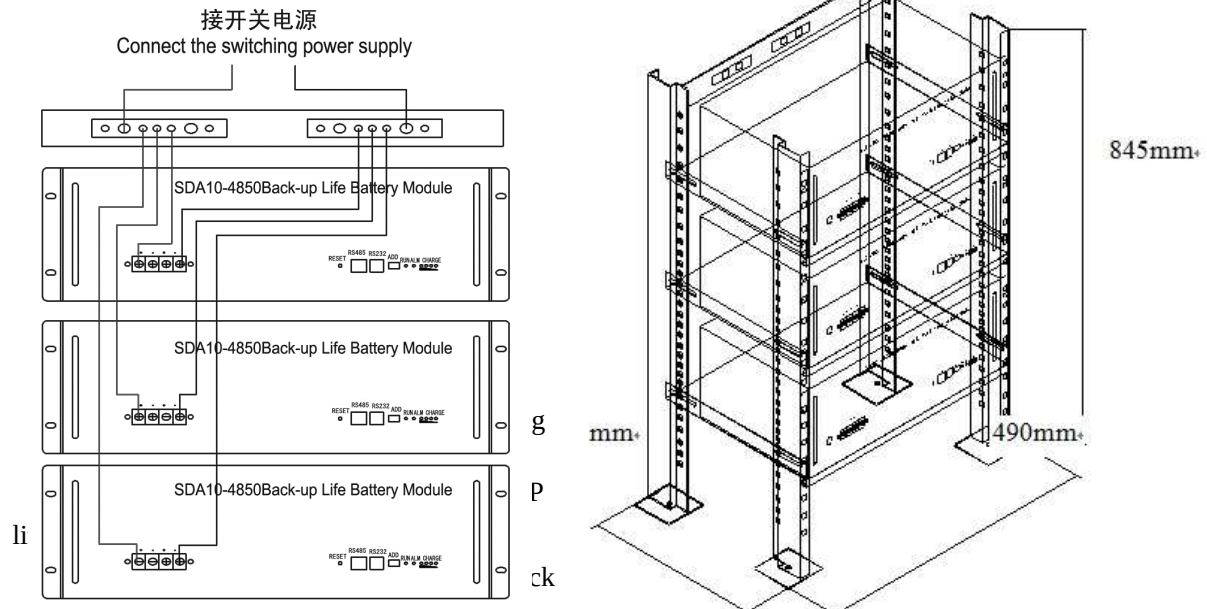
Picture 8-9 shows , use two red and black 10.0mm² flexible wires to connect the each LFP system's output "+" "-" terminals to the "+" "-" binding post's on the confluence copper bar , and consist a 100Ah LFP battery system.



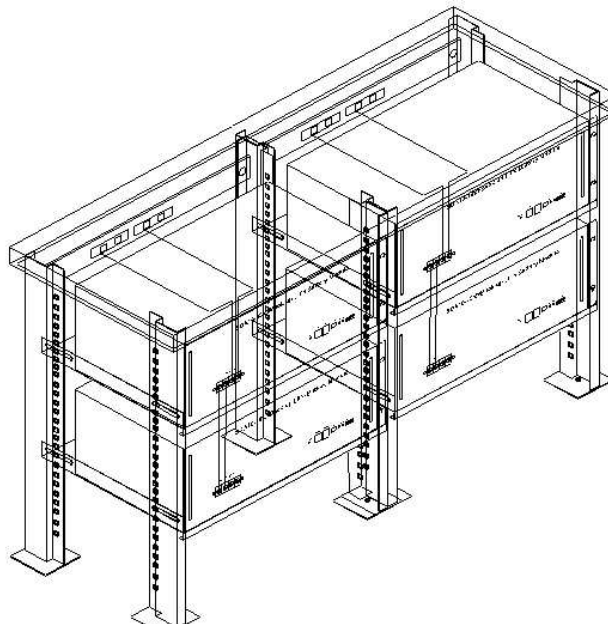
Picture 8-10 is a sketch map of the 100Ah LFP lithium communication battery

(3) The installation of scaffolding 150AhLFP communication battery system

Picture 8-11 showing , use two red and black 10.0mm² flexible wires to connect the “+” “-” output terminals of the three 50Ah LFP batteries to the “+”“-” terminals in holder’s confluence copper bar , and consist a 150Ah LFP lithium system.

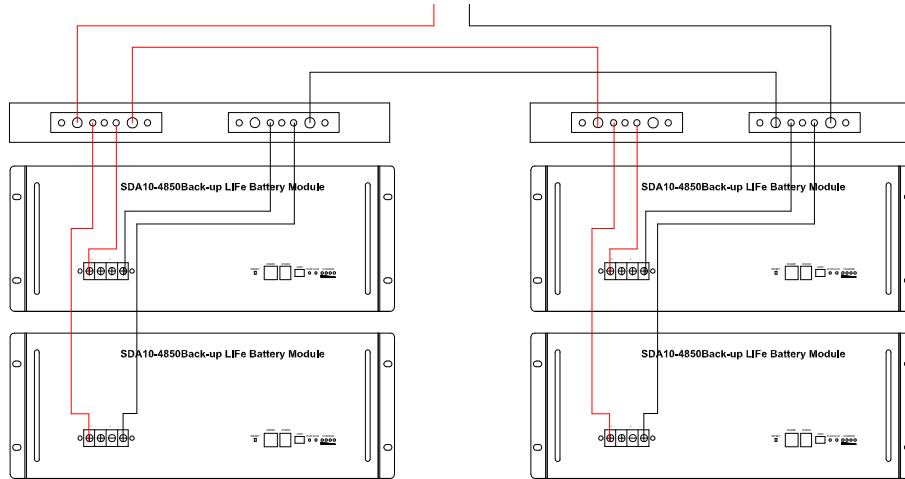


50.0mm² flexible wires to connect the 100Ah LFP system’s holder’s connection copper bar’s “+”“-” to another 100Ah LFP system’s holder’s confluence copper bar bar’s “+”“-”, and consist a 200Ah LFP lithium battery system.



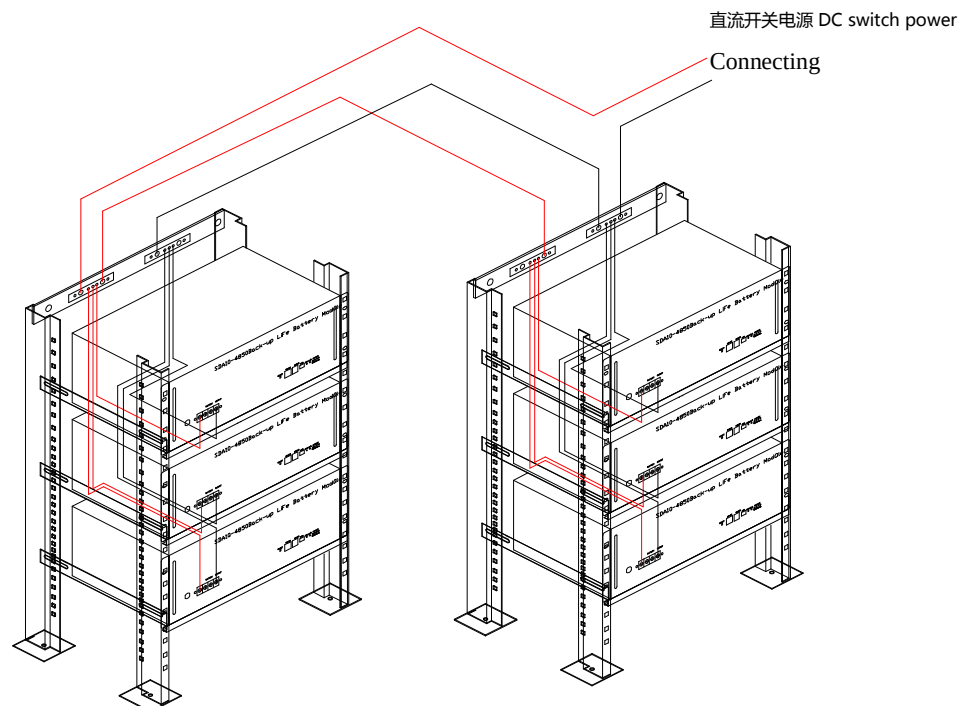
Picture 8-12 is a sketch map of installing a 200Ah LFP lithium battery

Connect the switching power supply

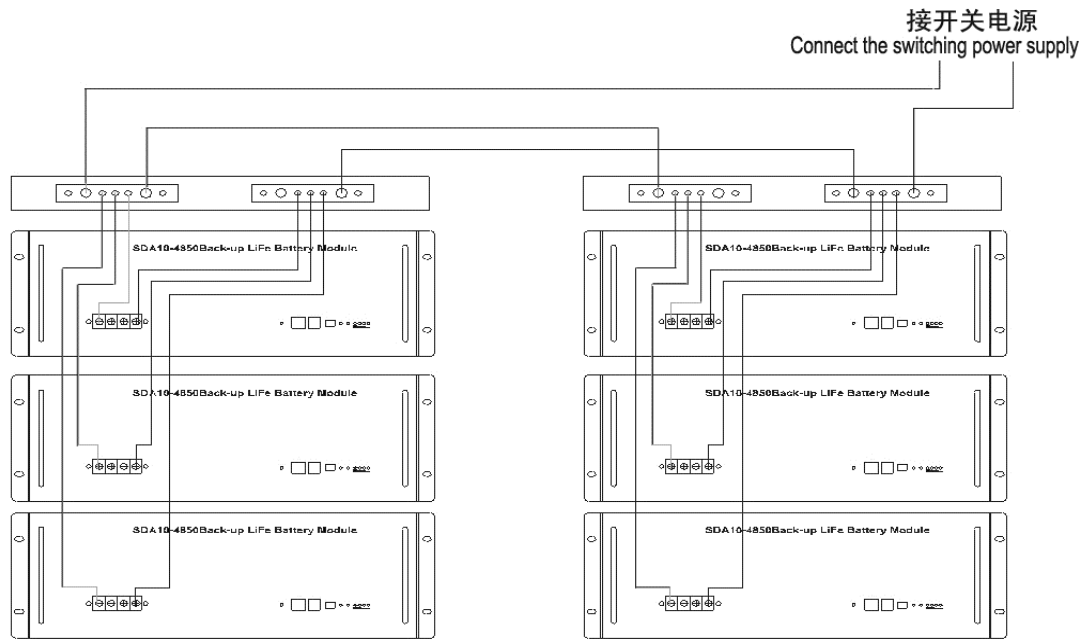


Picture 8-13 is a sketch map of installing a 200Ah LFP lithium battery
(5) The installation of a HSNF 300Ah LFP communication battery.

As picture 8-14、8-15 showing , use two red and black 70.0mm² flexible wires to connect a 150Ah LFP system's holder's copper bar's “+”“-” terminals to another 150Ah LFP lithium battery system's holder's confluence copper bar's “+”“-” terminals, and consist a 300Ah LFP lithium battery system.



Picture 8-14 is a sketch map of installing 300Ah LFP communication battery



8.1.3 Installment of equipment cabinet

Installment of equipment cabinet is a kind of indoor battery system which designed for the high demand of dust and water prevent situation. the system consist of a 19 inches standard equipment cabinet, battery pack, bus-bar and connecting wire etc. the height of the 19 inches equipment cabinet is designed by the number of the batteryys. The steps of installment are just below :

- (1) use two red and black more than 10 mm² flexible wires to connect battery system's +、- electrodes to the cabinet's confluence copper bars which connection in parallel , red line connect "+" ,and black line connect "-" ;
- (2) Arrange every battery modules on the equipment cabnet tray, use M6 bolts to fix all the hangers of the battery modules on the holders of equipment cabinet ;
- (3) From the right ground connection hole of battery system's back panel use a 2.5mm yellow green flexible wires,and make sure connection well ;
- (4) Outlet two lines of total positive and total negative from the cabinet's bus-bar .connect the lines to the switch power supply or equipment's positive and negative poles.

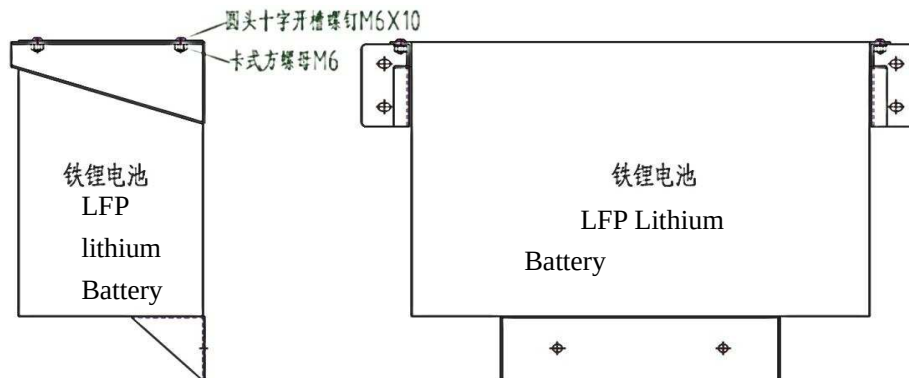


Installment of equipment cabinet

Installment of wall-hung type

(1) installment of battery system

As the picture 8-16 shows, the installment of the height 4U battery system is use four M8 expansion screws to fix battery's two hangers on the wall, the distance between them is 445mm, use two M8 expansion screws to fix the tripod under the battery, to support the battery.

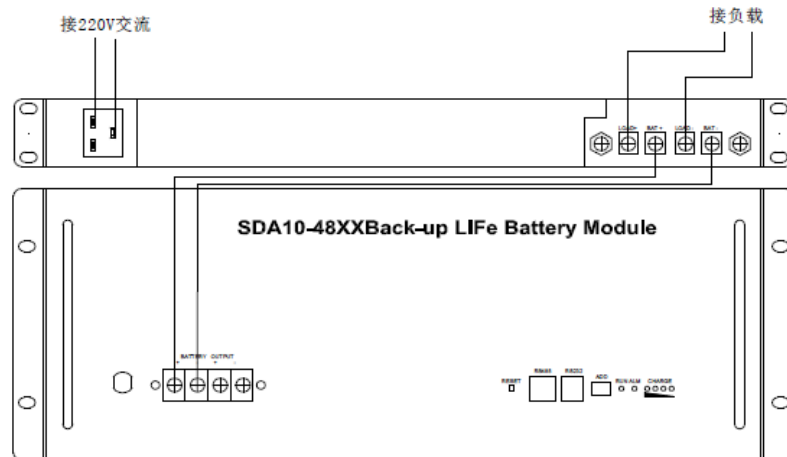


(2) Installment of dc unity

As the picture 8-18 shows, use four M8 expansion screws to fix the 5U parts on the wall, the distance between them is 445mm, hang 1U switch power supply and 4U battery system on the hanger's bayonet nuts with M6 bolts, battery and switch power supply panel all upwards, then use two M8 expansion screws to fix the tripod under them. As picture 8-18 and 8-19 shows.



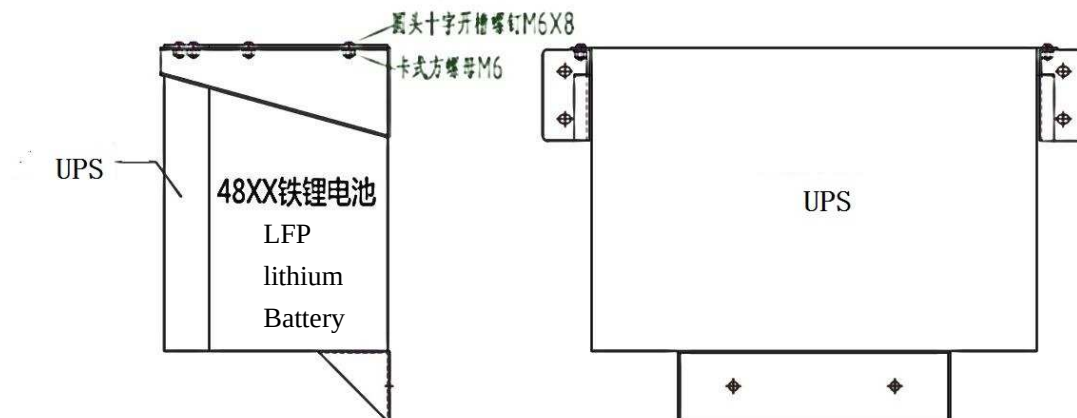
Picture 8-18 sketch map of battery and DC switch install on the wall
Battery and DC switch power supply wiring method as shown in figure 8-19:



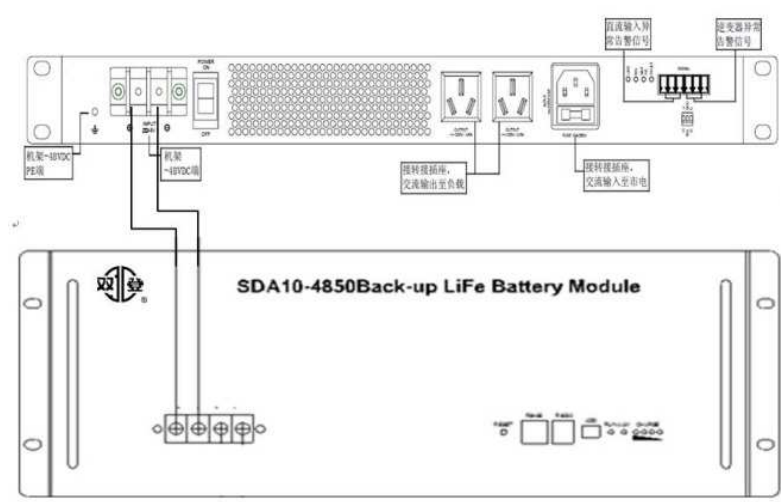
Picture 8-19 sketch map of battery and DC switch power-supply wiring

Installment of ac unity

As picture 8-20 shows ,use four M8 expansion screws to fix the hang wall component's hangers on the wall,the distance between them is 445mm,then fix the UPS and battery system on the hanger's bayonet nuts with M6 bolts, battery and ups panels all upwards, then use two M8 expansion screws to fix the tripod under them. 8-20 shows the sketch map of UPS hang on the wall..



Picture 8-20 The sketch map of battery and UPS hang on the wall
Battery and UPS wiring method as shown in figure 8-21:



Picture 8-21 sketch map of battery and UPS wiring

Comment : After the installing the battery system, DC integration, AC integration hang on the wall, We should cap dust covers above the panels, they can prevent the dust from obstructing equipment's fan and exhaust opening ,after working a long time .

8.2 System power on operation

When system installment finished it's in dormant mode, when need to use it, just to do a simple power on operation, then the equipment can work normally.

8.2.1 Activate system

During storage and transportation , or with no external power source and non-load 24 hours later, the battery system are in the dormant mode. after with the 48V voltage , the battery system be activated, turn into the normal working mode, can be charged and discharged, or in the waiting mode. What must be specially explain is that if the battery system in the dormant mode , the DC power switch must off its reverse connect protection function, after AC power recover, the 48V voltage will activate the battery system

8.2.2 System standby

When the system activated, if off its external power source and load, battery system will initiative into the ready mode. under this mode, power lose very less, the run light shining shows the battery system are in ready mode. after 24 hours, the battery system automatic into dormant mode. Under this mode, power lose more less, all the lights in the panel are blackout , means the battery system in dormant mode.

8.3 Alarm instruction and solution

When system breakdown, the system will give the alarm signal as the breakdown situation, all the breakdowns will have acousto-optic warning and letter warning. When the system alarm , far-end monitoring center will get the warning signal, then the maintance worker should check the equipment by the signal from the State Monitoring Unit immediately, and find out the fault type and fault place, give the right solution..

8.3.1 The alarm and solution for the fault which affect the system's output

When the single cells in the system appear some faults, such as over charge protection, over current protection, low voltage protection, temperature protection and so on, please according to the solution in chart 8-1

Chart 8-1 alarm and solution

Mode	Alarm value	The warning instructed	Processing
Charge mode	Charging abnormal	ALM off	Stop charge , Search for fault reason
	Charge over current	ALM bright	Stop charge , Search breakdown reason
	Charge temperature protection	ALM bright	Stop charge
Discharge mode	Discharge over current protection	ALM bright	Stop discharge , Search for fault reason
	Discharge temperature protection	ALM bright	Stop discharge
	Total voltage low protection	ALM off	Charge
	Cell low voltage protection	ALM off	Charge

8.3.2 Alarm and solution which have no effect on the system's output

When total voltage low or single cell voltage low the battery system will show alarm signal. The maintenance worker must check the equipment according to the alarm message, find out the fault and position ,then give the right solution to make sure the whole system in good working situation. Avoid the fault amplify to affect system's output. The phenomenon and solution as the chart 8-2 below..

Chart 8-2 Alarm but no protection

Alarm volume	Mos -FET action	Alarm signal	Solution
Total low voltage alarm	No	ALM shinning	Stop discharge
Cell voltage low alarm	No	ALM shinning	Stop discharge

8.4 Communication fault

From the software in the host computer we can see the protection mode , system mode , which can help us to find out the fault reason correctly.

- (1) When RS232 appear communication fault , we can solve it according to following steps:
 - a.Check communication circuit;

- b.Reset MasterPack;
 - c. For the cascade system, we can exchange the MasterPack setting and SlavePack setting to diagnose
- (2) When RS485 occur communication breakdown,according to following step elimination:
- a. Check communication circuit;
 - b. Reset SlavePack;
 - c. Exchange slave pack address to diagnose.

8.5 Protection fault

8.5.1 Over voltage protection

When use the standard charge voltage, usually will not occur internal cell over voltage, only if between the cells have a big inbalanced. When one cell over voltage but total voltage low, the protection will happen. Need check the cells' voltage.

8.5.2 Current protection

When the system's current small, and occur current protection, need to check the status indication,check the connection between cells is normal or not.

8.5.3 Other kind of fault

If the battery system occur protection alarm and can not be charge and discharge, we need to check voltage, temperature and system condition instruction, find out the fault's reason

8.6 Solution for special circumstance

8.6.1 Power off

AC no power is the most common thing when the system is working . when the power off time is short, AC supply can be supported by the LIP lithium battery in the system. If the power off time is not clear or very long, we should care about the system's power-on time.

8.6.2 Cataclysm accident

Cataclysm accident are include:thunder stroke, flood, earthquake, fire and other disasters which can lead the communication facility's fault. For the disasters which affect upon the communication security should be prevented first, at the same time, the communication site should have manpower and material resources to deal with these disasters, and the communication site should have emergency state management rules and major accident urgent repair rules.

9、 Packing transportation and storing

9.1 Packing

The system should be whole packed, to make sure the product will not get any damage from noxious gas, chemistry pollution, static electricity, moist and the mechanical during transporting, storing. Packing cases should be indicated : product name, type and specification ,production date, quantity and batch number.

9.2 Transportation

Adapt to the transportation of vehicle , ship ,and flight, during transporting should cover the product, avoid insolation, handle with care. Falling, turn over, or pressing are prohibited. During transporting, should avoid rain, snow and mechanical impact.

9.3 Storing

When system storing should place in the dry warehouse, no insolation and the rain drenches. Harmful gas, inflammable, explosive and corrosive chemicals are not allowed in the warehouse, to avoid mechanical shock, stress, and strong magnetic field, avoid direct sunlight and the distance between the the heat source can't less than 2m, The box should be above the ground at least 20cm, the distance between wall, window or air inlet at least 50cm.**After storing more than 3 months must recharge the battery. After storing more than 6 months must do capacity test, if qualified , can be used.**

10、 Environment protection request

Satisfy RoHS authentication request , no cadmium、hydride、fluoride or asbestos , never send out organic compound , wrappage can be recycle.

11、 Safe matters needing attention

- (1) Can't immerse the battery in water, during storing, put the battery in dry and cool environment;
- (2) Can't store or use the battery near high temperature or high heat source place;
- (3)Please use the system according to the stipulates and parameter in the user's manual;
- (4) Can't connect the system to the power socket directly;
- (5)Can't discard the battery into fire or heater;
- (6) Can't break up the battery and the parts;
- (7) Can't rap, steps on or throw the battery ;
- (8)When AC power off, this system's output terminal still have the voltage, when the use the battery should look out electric shock or short-circuit;
- (9)Please read the product manual carefully before using.

(10) Installation and debugging before the battery, please first to grounding cable, ensure good grounding!

Note: a full range of product model, please pay attention to product sheet specifications, user manuals will not be updated regularly, update time manufacturers decide!